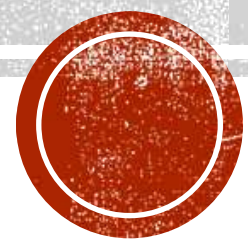
GFN1000
IN DIALOGUE WITH NATURE
與自然對話

Darwin, *On the Origin of Species*



Content of the Course

Physical Universe

- Plato, Lindberg
- Aristotle, Lindberg
- Cohen, Newton

World of Life

- Darwin
- Watson
- Carson

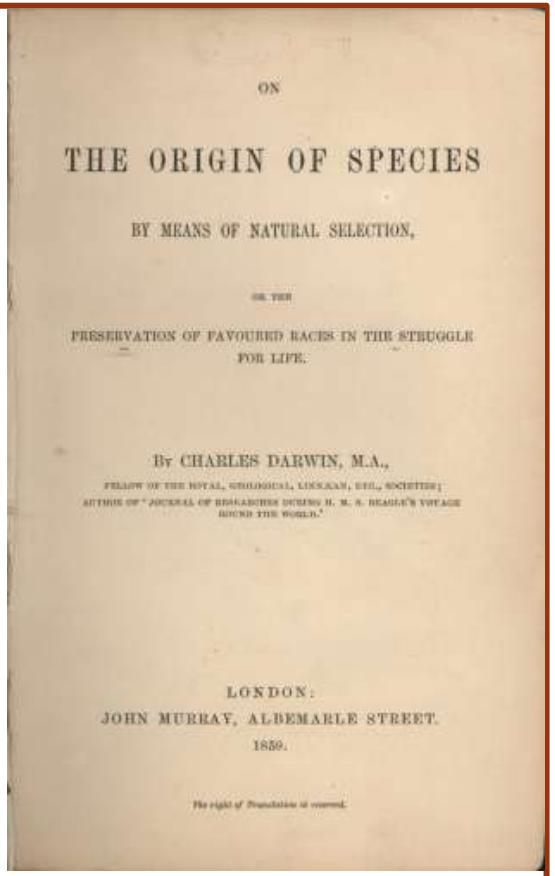
Human Understanding

- Poincare
- Kandel
- Needham
- Sivin, Shen Kuo
- Dunhan, Euclid

Core Question

What are the laws of life?





ON

THE ORIGIN OF SPECIES

BY MEANS OF NATURAL SELECTION,

OR THE

PRESERVATION OF FAVOURED RACES IN THE STRUGGLE FOR LIFE.

By CHARLES DARWIN, M.A.,

FELLOW OF THE ROYAL, GEOLOGICAL, LINNÆAN, ETC., SOCIETIES;
AUTHOR OF "JOURNAL OF RESEARCHES DURING H. M. S. BEAGLE'S VOYAGE ROUND THE WORLD."

LONDON:
JOHN MURRAY, ALBEMARLE STREET.
1859.

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TODAY'S FOCUS

- Background
- Evolution & natural selection
- Public responses



Charles Darwin

- English naturalist and geologist
- Charles Darwin grew up in a wealthy family.
  - Father (Robert): physician
  - Paternal grandpa (Erasmus Darwin): physician, natural philosopher, and poet
  - Maternal grandpa (Josiah Wedgwood): founder of Wedgwood
- He studied medicine (Medical School, University of Edinburgh, 1825) initially, then theology (Cambridge University, 1828), then became a naturalist as a student of botanist and geologist J. S. Henslow.



Charles Darwin
(1809-1882)

He was crazy for beetles …



← A cartoon of Darwin

Darwin's → beetle box



I will give a proof of my zeal: one day, on tearing off some old bark, I saw two rare beetles and seized one in each hand; then I saw a third and new kind, which I could not bear to lose, so that I popped the one which I held in my right hand into my mouth. Alas it ejected some intensely acrid fluid, which burnt my tongue so that I was forced to spit the beetle out, which was lost, as well as the third one.

The Autobiography of Charles Darwin 1809-1882

Charles Darwin

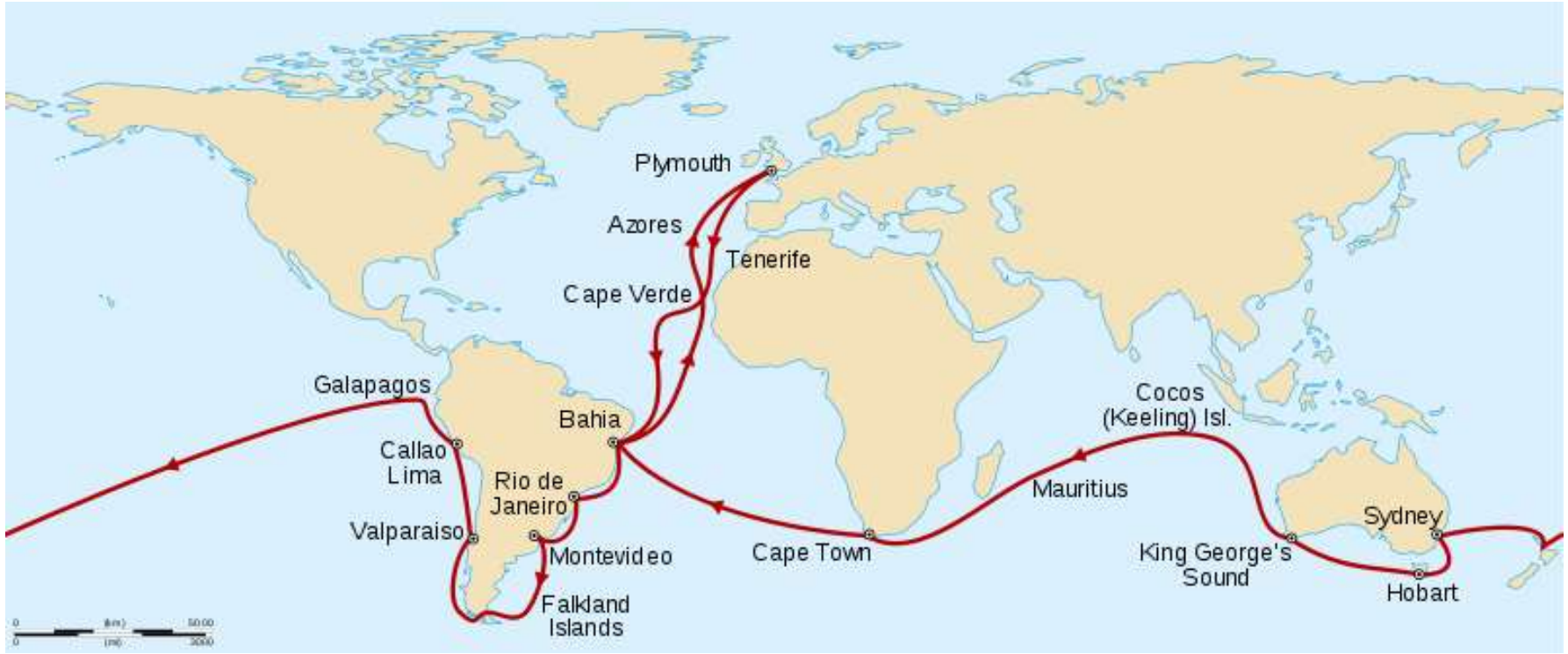
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- Author of *On The Origin of Species; The Descent of Man*
- Proposed evolution by natural selection (changing science forever)
- Buried in Westminster Abbey, and became neighbor with Newton



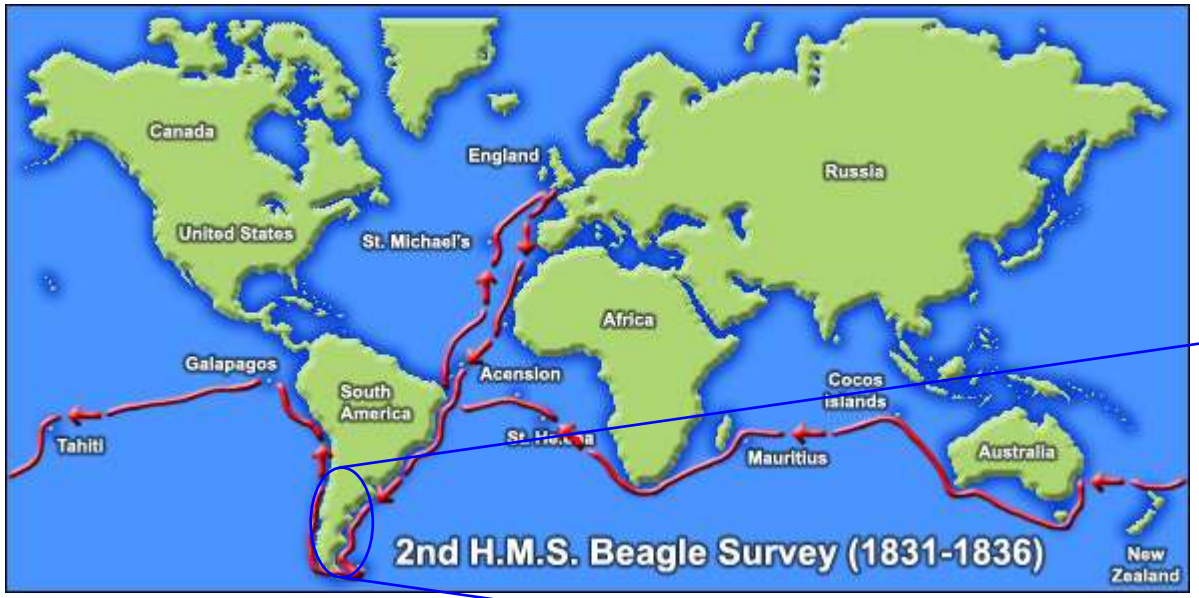
Charles Darwin
(1809-1882)

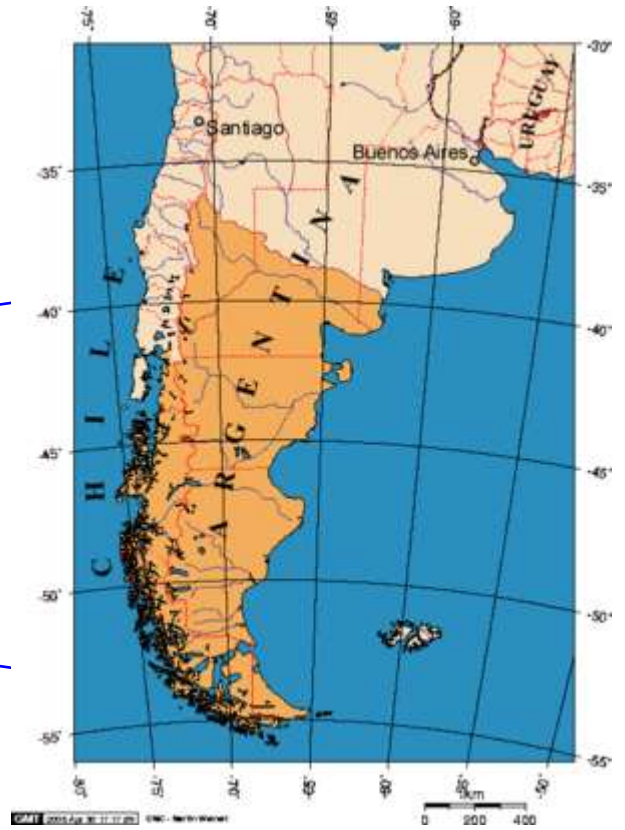
Where Did Darwin's Idea Come From?

- Darwin travelled on a British warship, HMS *Beagle* for a 5-year voyage (Dec 1831- Oct 1836) around the world.
- In these 5 years, he was on land for 2/3 of the time and performed geological and zoological observations.



An Adventurous Journey-HMS Beagle

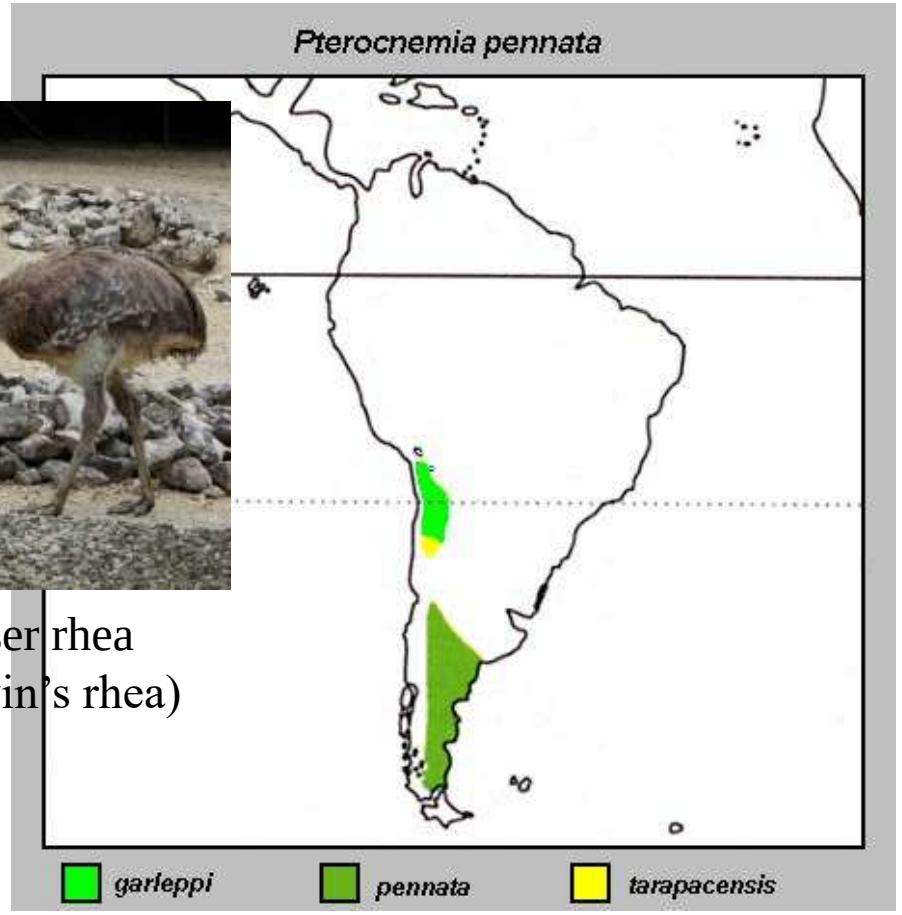




DARWIN'S RHEAS (小美洲鸵)

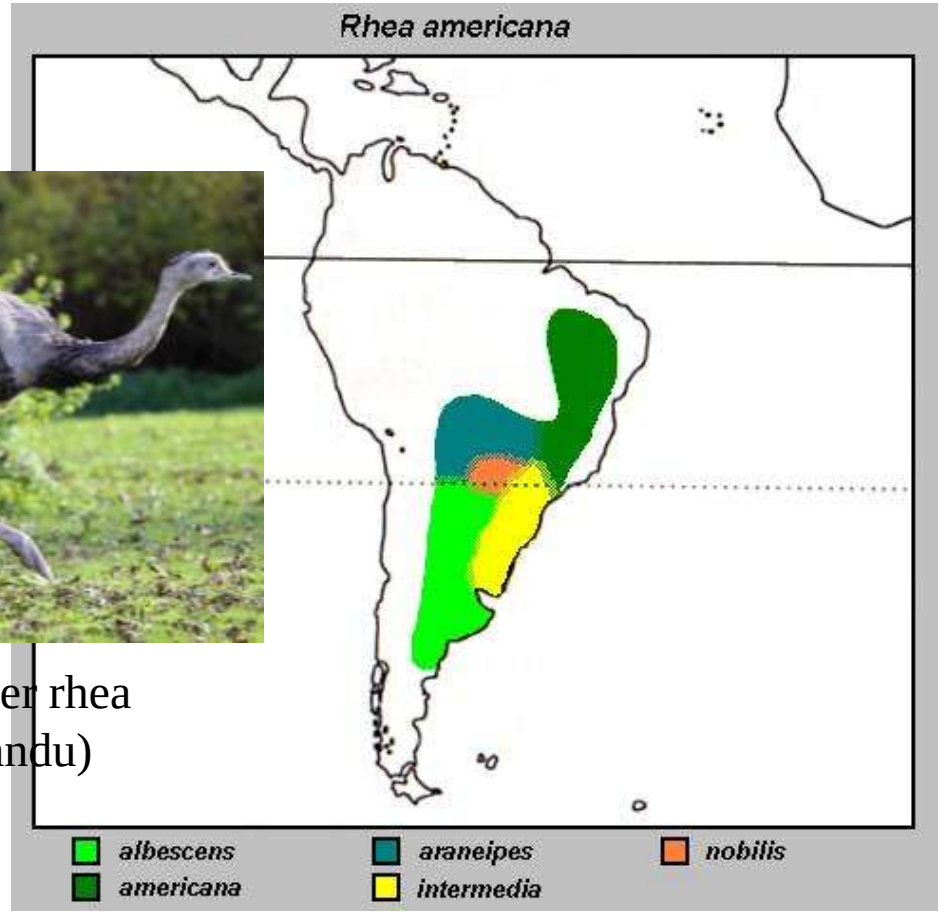


Lesser rhea
(Darwin's rhea)

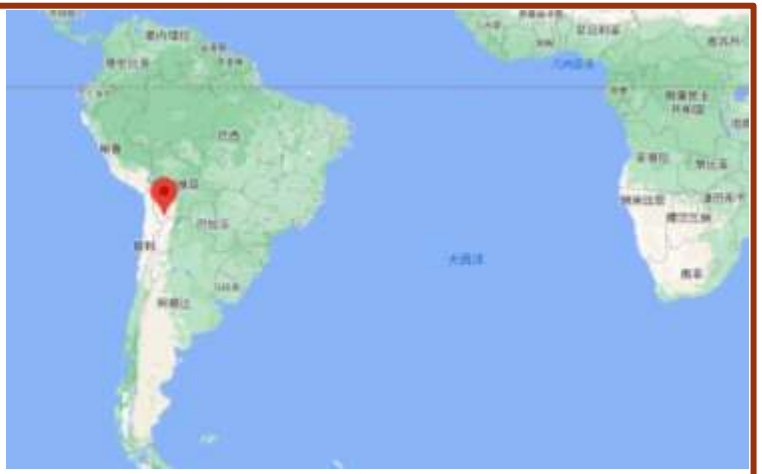


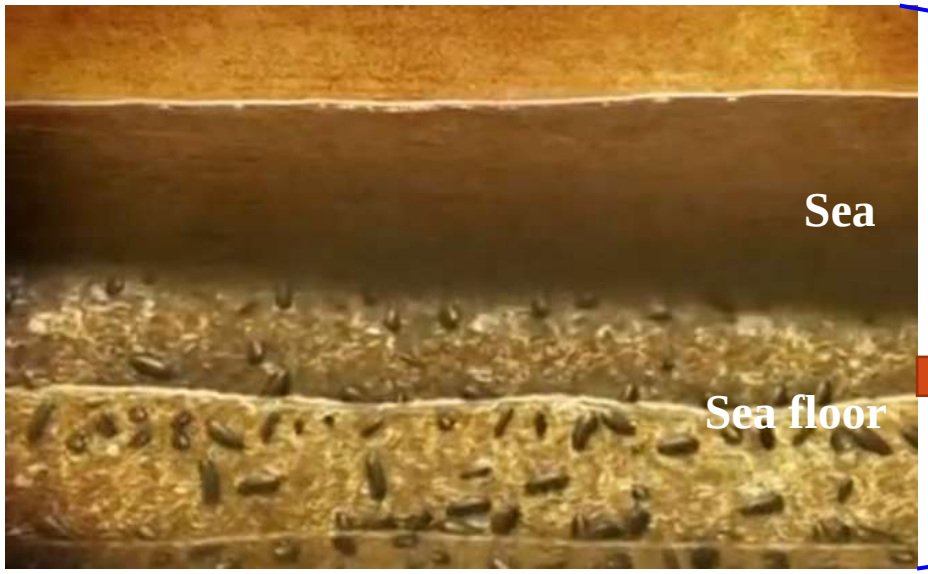


Greater rhea
(Nandu)

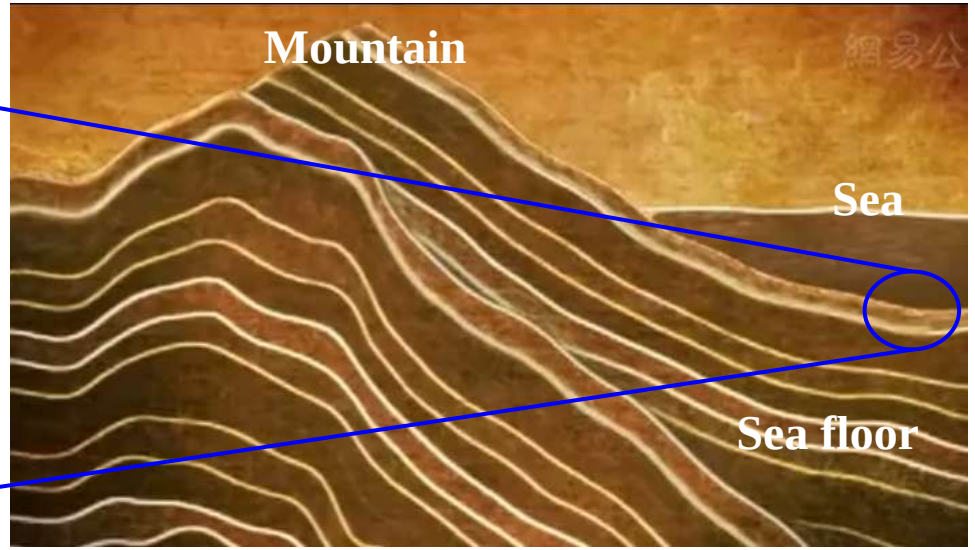


FOSSILS ON THE ANDES

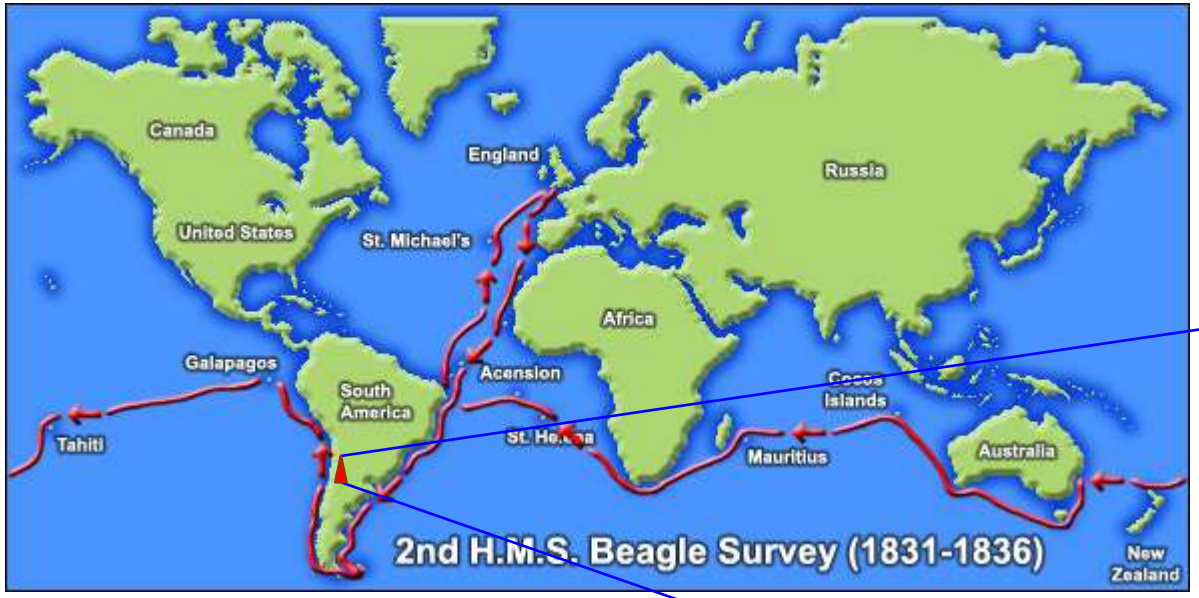




Immense forces drove the layers upward thousands of feet from where it started



AN EARTHQUAKE IN CHILE





Charles Lyell's *Principles Of Geology*



(1797-1875)
English geologist

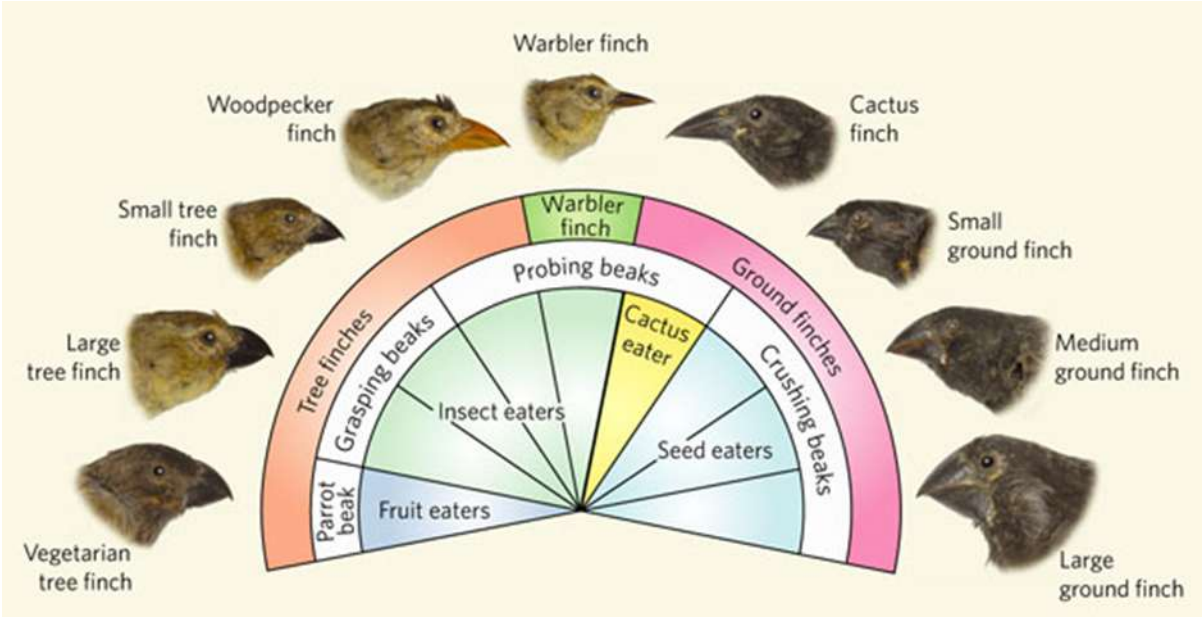
- Argued against catastrophism (灾变说)
- The Earth was shaped by slow-moving forces still in operation today.
- Geologic change as the steady accumulation of minute changes over enormously long spans of time.
- Fossils present evidence for gradualism (渐进主义)



Galapagos Islands: The Beak Of The Finch

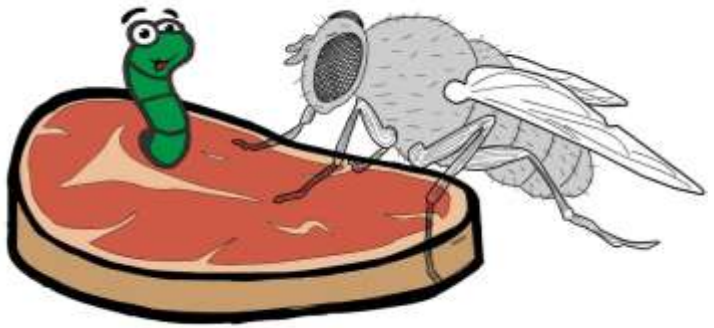
- Found many varieties of finch that were on different islands
- The beaks of the finch variance due to different food sources





DARWIN BECAME SUSPICIOUS

- Critical of the Bible's story of creationism.
- Aristotle's "spontaneous generation"
- Lamarck's "innate life force"
- …



Thomas Malthus' *The Principle Of Population* (人口论)



Thomas Malthus
(1766-1834)
English scholar

- Malthus observed that <u>human populations</u> can increase in size faster than their ability to produce food.
  → Overpopulation
  → Struggle for existence
- Inspired Darwin's natural selection

Darwin: applied it to the whole world of life?

Emergence of the idea of evolution theory

Observations

- Populations increase exponentially (reproduction)
- Finite capacity to support a population in a region
- Variations in fitness of members in a population

Deductions

- Competition among populations
- Fittest variants will survive
- Adaptive evolution

But Darwin did not publish his findings until…1859



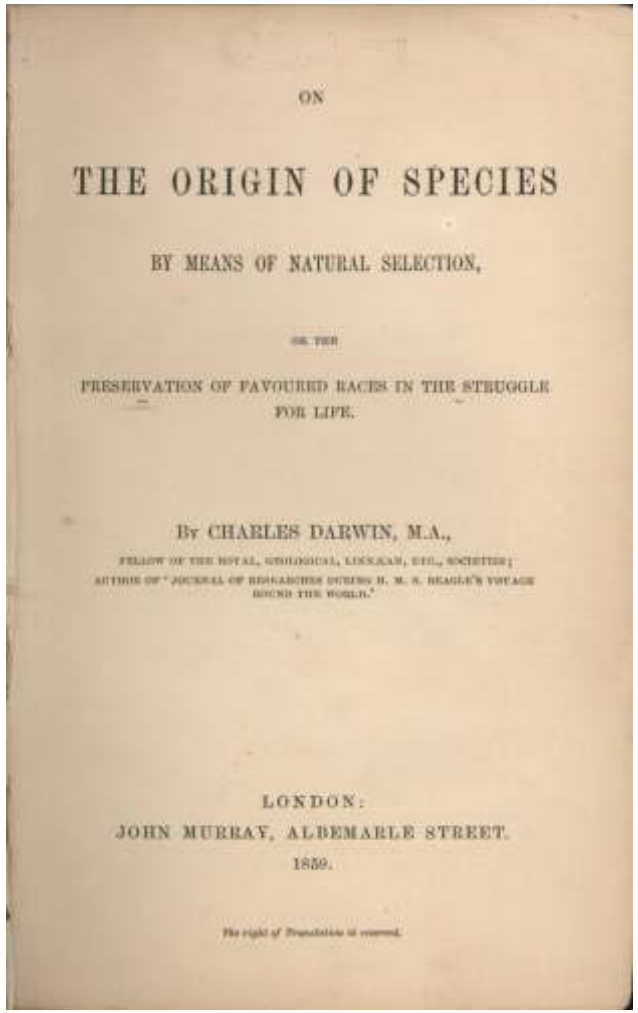
Alfred Russell Wallace

- English naturalist, explorer, social activist
- In 1858, Wallace sent his essay to Darwin
- Theory similar to Darwin's (Darwinism could have been Wallacianism).
- Wallace's theory was published, along with a description of Darwin's own theory, in 1858.
- Forced Darwin to publish his 490-page "abstract" in 1859.



(1823-1913)

THE ORIGIN OF SPECIES, 1859



ON

THE ORIGIN OF SPECIES

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Content of the book:

Chapter I Variation Under Domestication
Chapter II Variation Under Nature
Chapter III Struggle for Existence
Chapter IV Natural Selection
Chapter V Laws of Variation
…

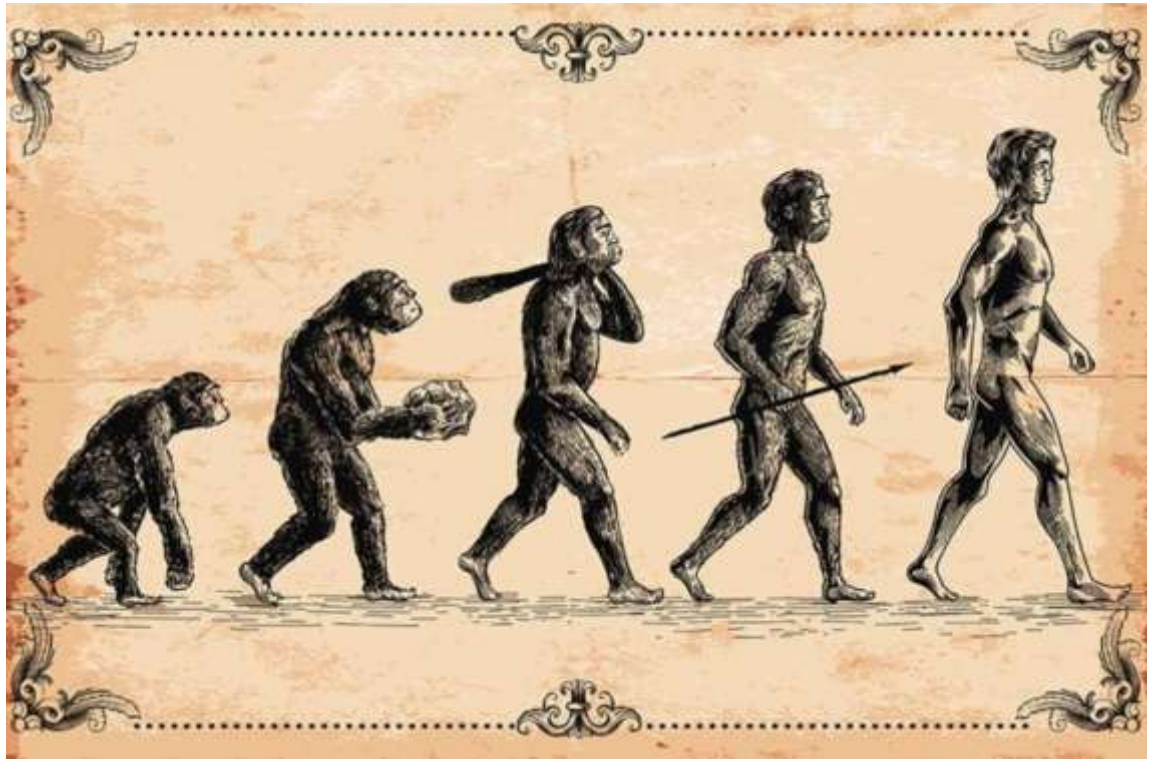
TODAY'S FOCUS

- Background
- Evolution & natural selection
- Public responses



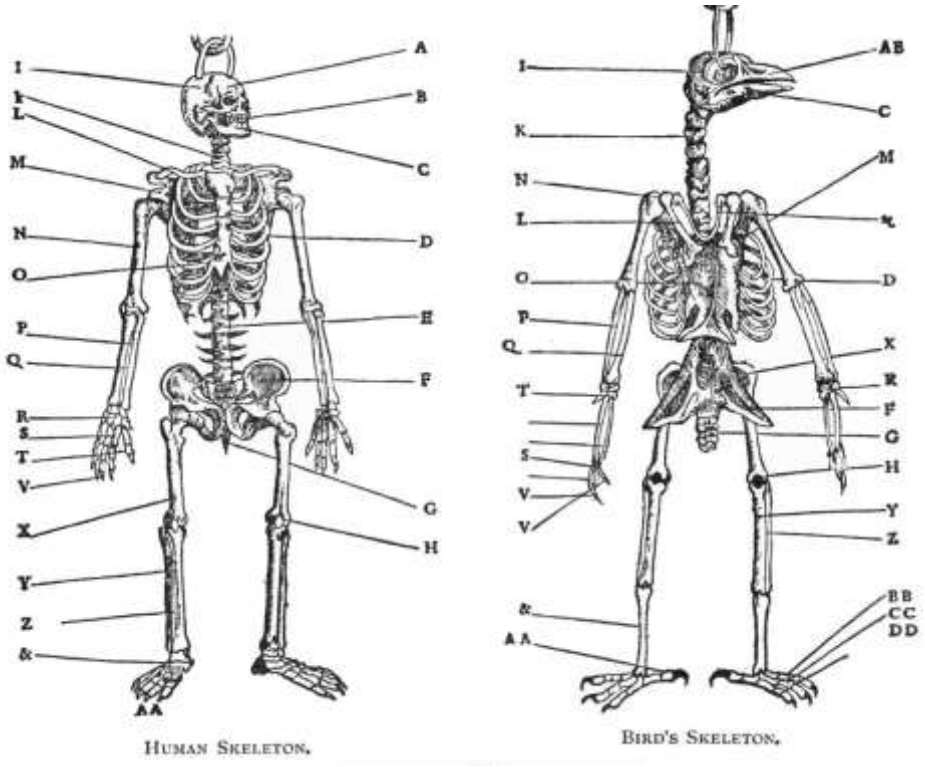
WHAT IS EVOLUTION?

- Evolution today is defined as the change in heritable characteristics of biological populations over successive generations.



EVOLUTION EVIDENCE

1. Variation under domestication
2. Comparative anatomy



A comparison of the skeleton of birds and man in Belon's book on birds, 1555

3. Fossil record/Paleontology
4. Embryonic development
5. Biogeographical distribution (marsupials such as kangaroos)

…

Evolution Explanation

- Species could degenerate into different organisms.

  -Count George-Louis Leclerc de Buffon (1707-1788)

- All warm-blooded animals could have descended from a single filament.

  -Erasmus Darwin (1731-1802)

- "Law of use and disuse"

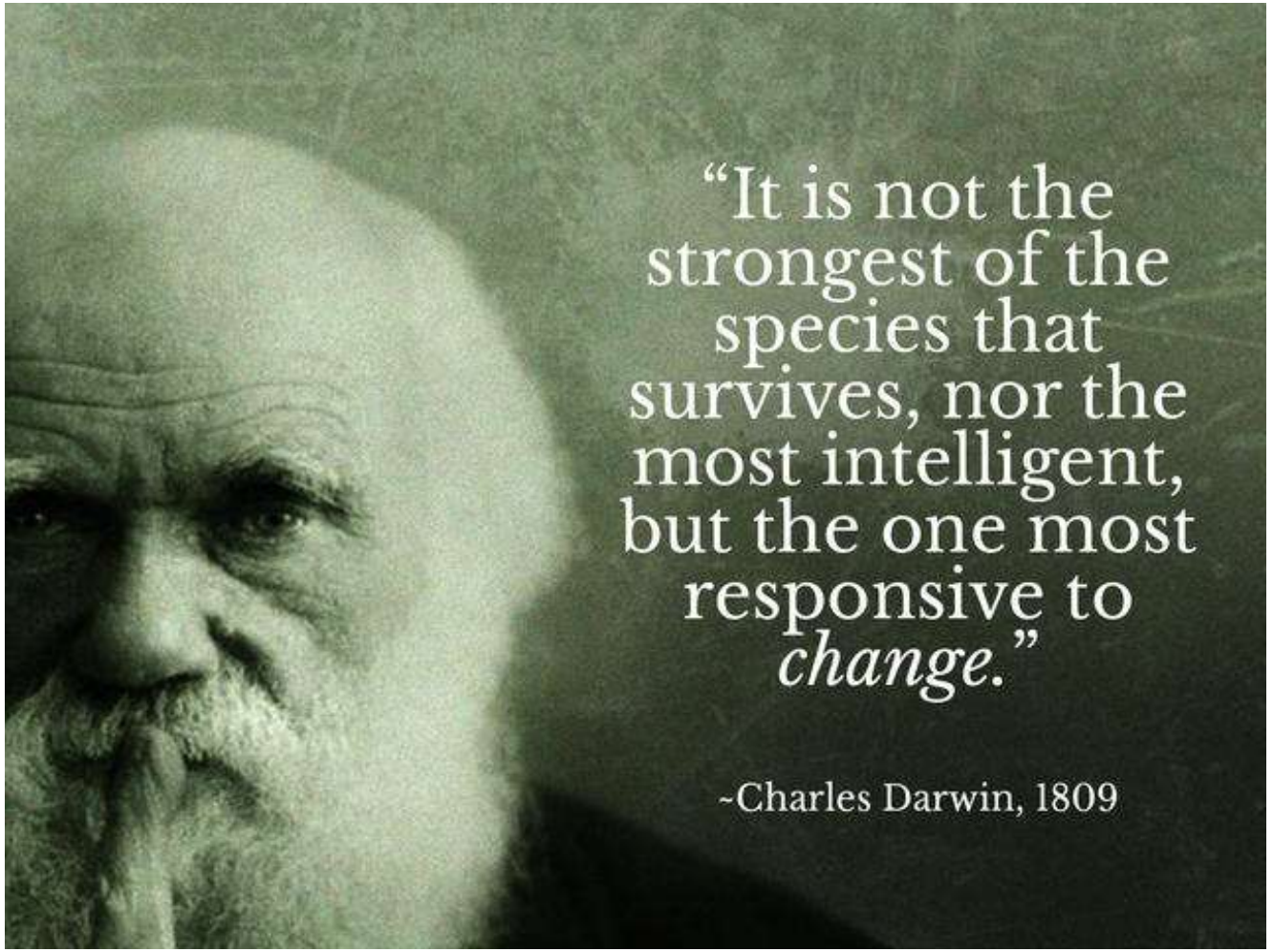
  -Lamarck (1744-1829)

- Geoffroy Saint-Hilaire (1772-1844)
- William Charles Wells 1813
- Patrick Mathew 1831
- …



Natural selection

- Darwin did not come up with the idea of evolution, but its mechanism.



Natural Selection (Summarized By Biologist Ernst Mayr)

Every species is fertile enough that if all offspring survived to reproduce, the population would grow.

Populations generally remain stable once they reach a certain size.

Resources such as food are limited.

A struggle for survival ensues

Struggle for Existence

- E.g. when a country undergoes some climate change, inhabitants would be greatly affected. Some species could migrate to find a better home.

- This will lead to **competition for food** between the foreign intruders and the original inhabitants. That is, the **struggle for existence.**

- The most **adaptable** species will be preserved.



Natural Selection (Summarized By Biologist Ernst Mayr)

Every species is fertile enough that if all offspring survived to reproduce the population would grow

Populations generally remain stable once they reach a certain size.

Resources such as food are limited.

A struggle for survival ensues

Individuals in a population vary significantly from one another

Much of this variation is heritable

Three Kinds Of Variations

- Favorable variations
- Injurious variations

→ **Natural selection**

- Variations neither useful nor injurious

Natural Selection (Summarized By Biologist Ernst Mayr)

Every species is fertile enough that if all offspring survived to reproduce the population would grow

Despite periodic fluctuations, populations remain roughly the same size

Resources such as food are limited.

A struggle for survival ensues

Individuals in a population vary significantly from one another

Much of this variation is heritable

Individuals more suited to the environment are more likely to survive and reproduce and leave their heritable traits to future generations.

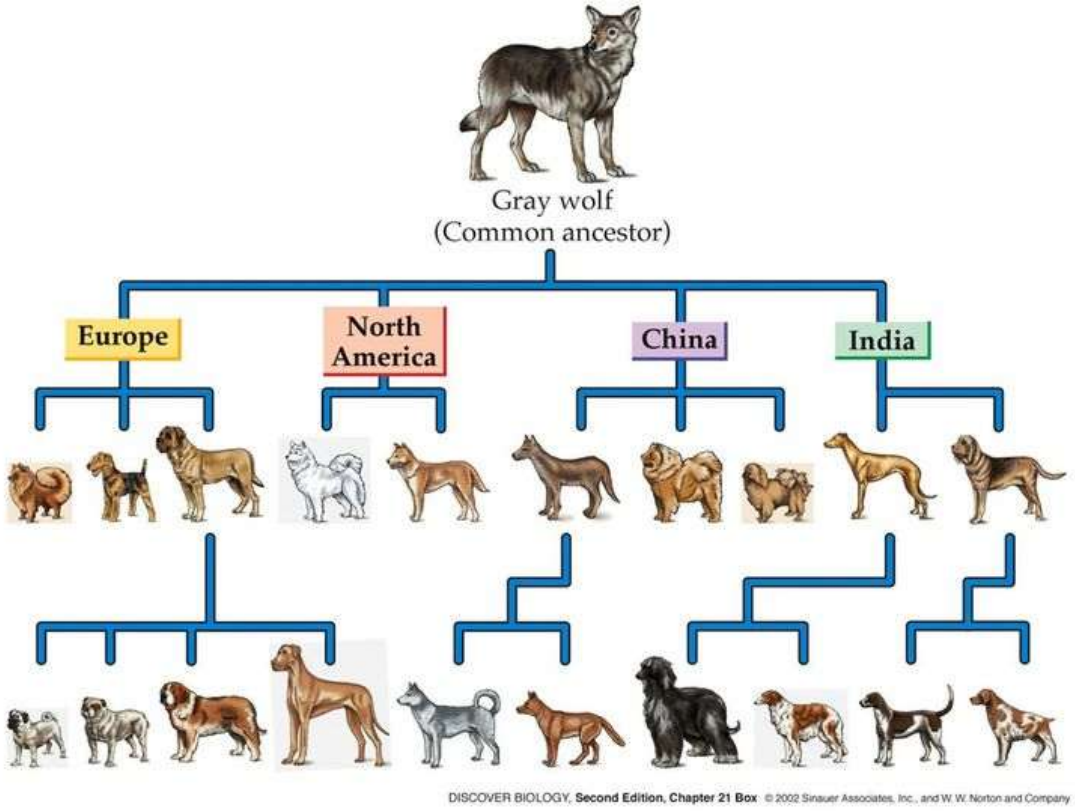
These variations accumulate over time to form new species.

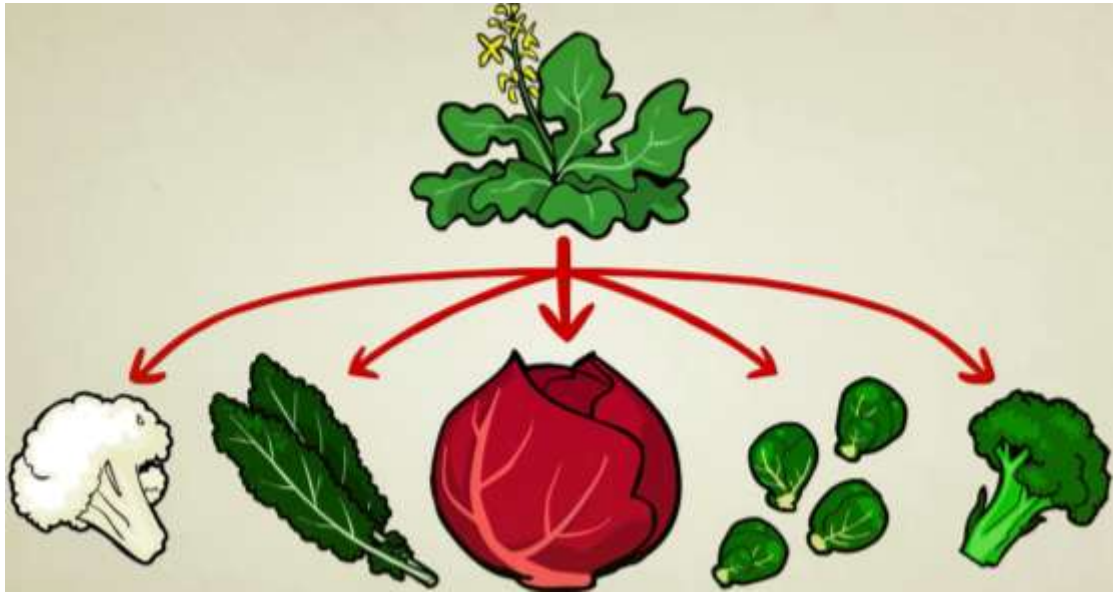


Artificial selection

Selection causes change!

Cause	Selection
Human	Artificial selection
Nature	Natural selection



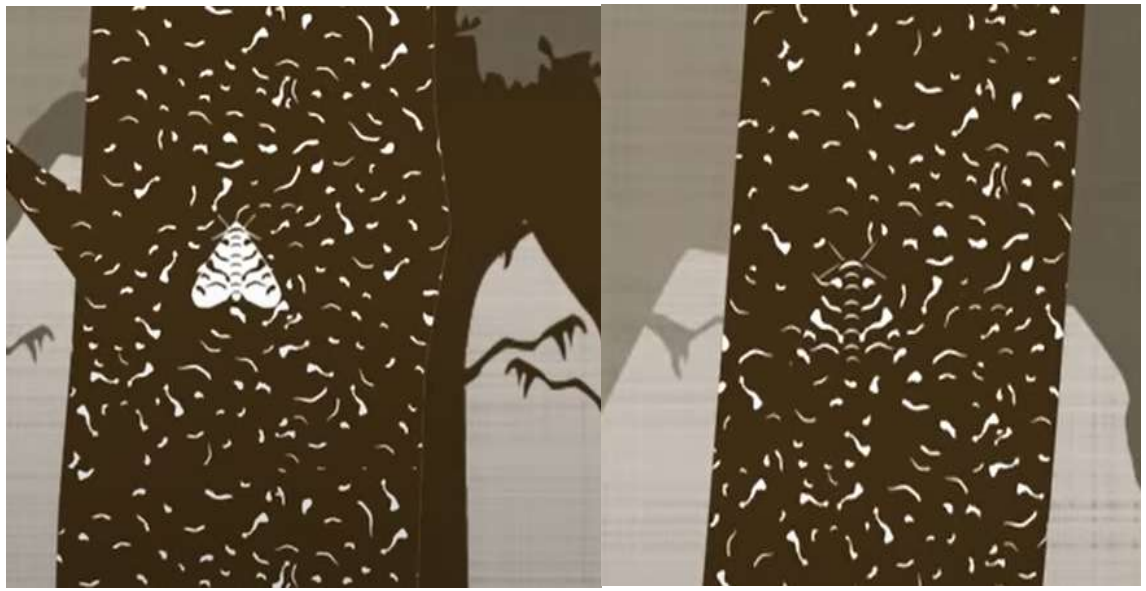


Human developed these different breeds by cultivating specific traits for long spans of time.

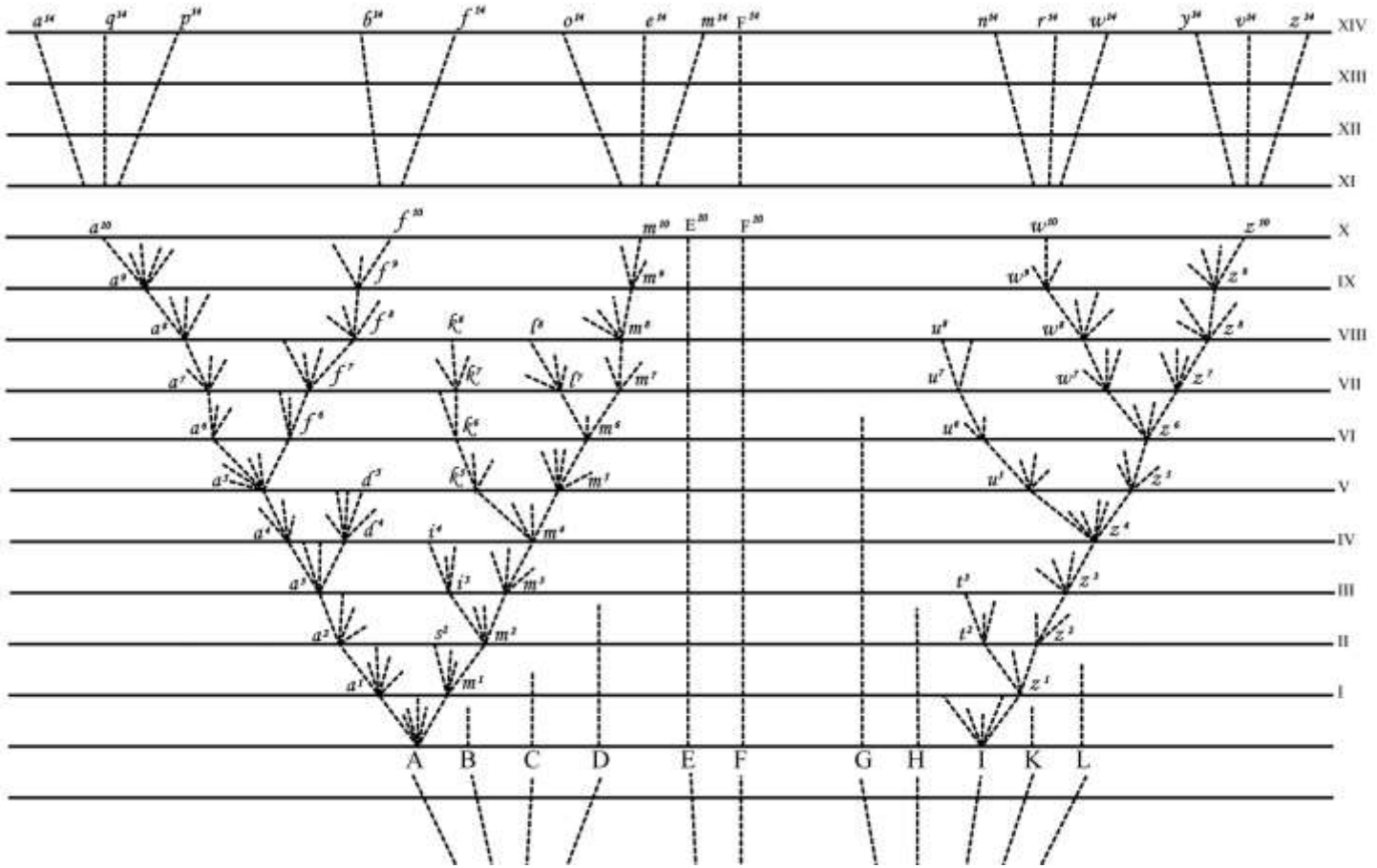
NATURAL SELECTION

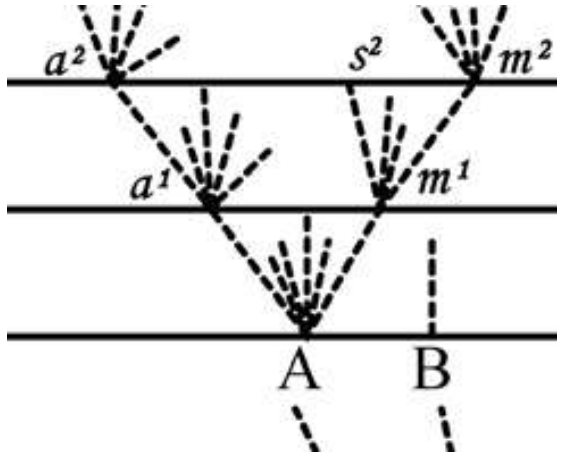
Darwin then suggested that nature could do the same kind of selection as human do via **environmental changes**. This he called **natural selection**.





The Diversity of Life on Earth: the Tree of Life

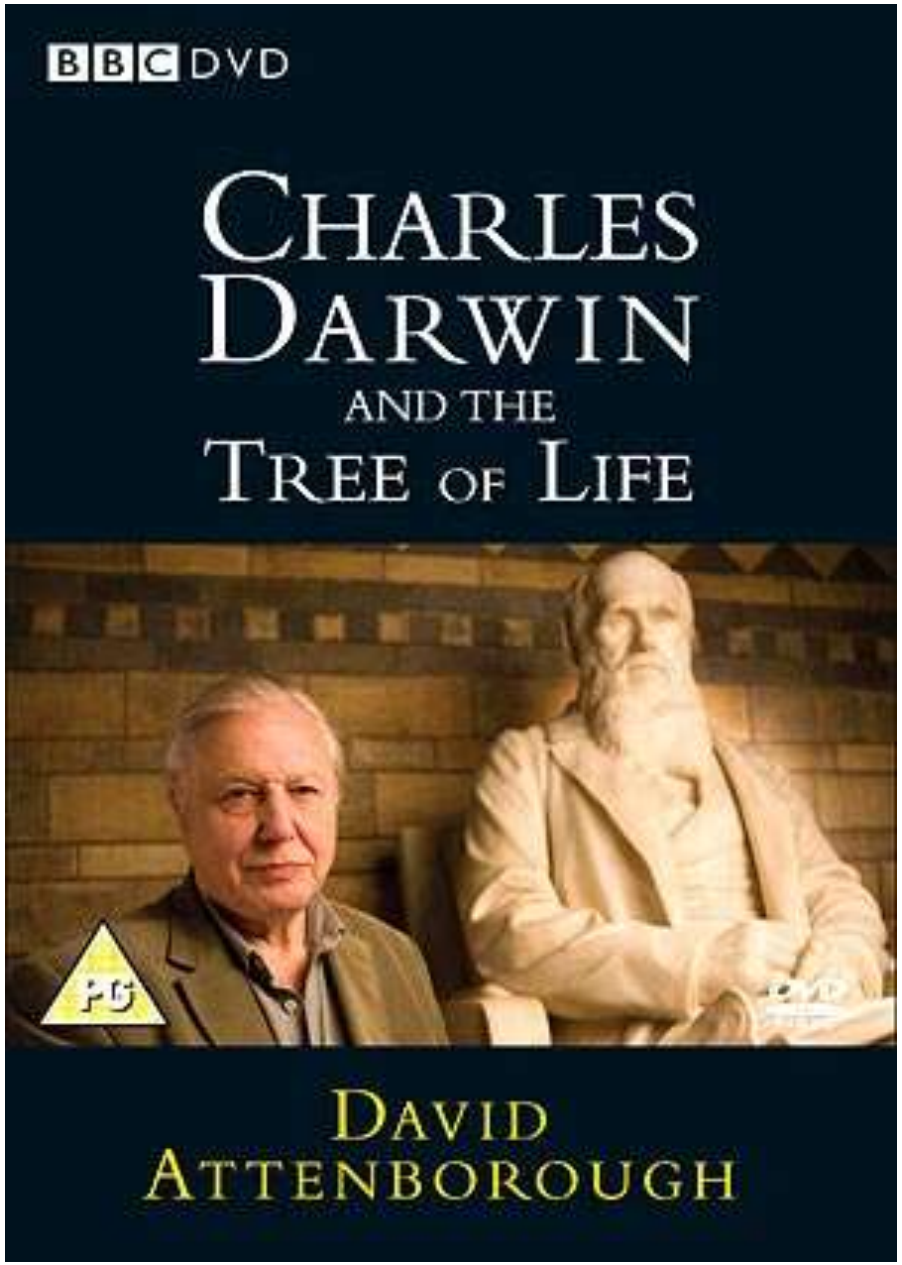




Taxonomy

种 species
属 genus
科 family
目 order
纲 class
门 phylum

"Thus the small differences distinguishing varieties of the same species, will steadily tend to increase till they come to equal the greater differences between species of the same genus, or even of distinct genera."



Year 2009

200th anniversary of the birth of Charles Darwin

150th anniversary of the publication of *On the Origin of Species*

CHARLES DARWIN AND THE TREE OF LIFE

"Current estimates of the number of different species vary from 6 million to 100 million."

"There are often a multitude of variations on a single pattern –

Nearly 200 different kinds of monkeys, for example.

And 315 humming birds.

Nearly a thousand bats.

And beetles at least 350,000 species of them."

TODAY'S FOCUS

- Background
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- Public responses



Laws of Nature

Newton's *Principia* (1687)

- The laws of motion (cf. Text 9 "mechanical worldview")

Darwin's *On the Origin of Species* (1859)

- The laws of life

- "There is grandeur in this view of life, with its several powers, having been originally breathed into a few forms or into one; and that, whilst this planet has gone cycling on according to the fixed law of gravity, from simple a beginning endless forms most beautiful and most wonderful have been, and are being, **evolved.**"



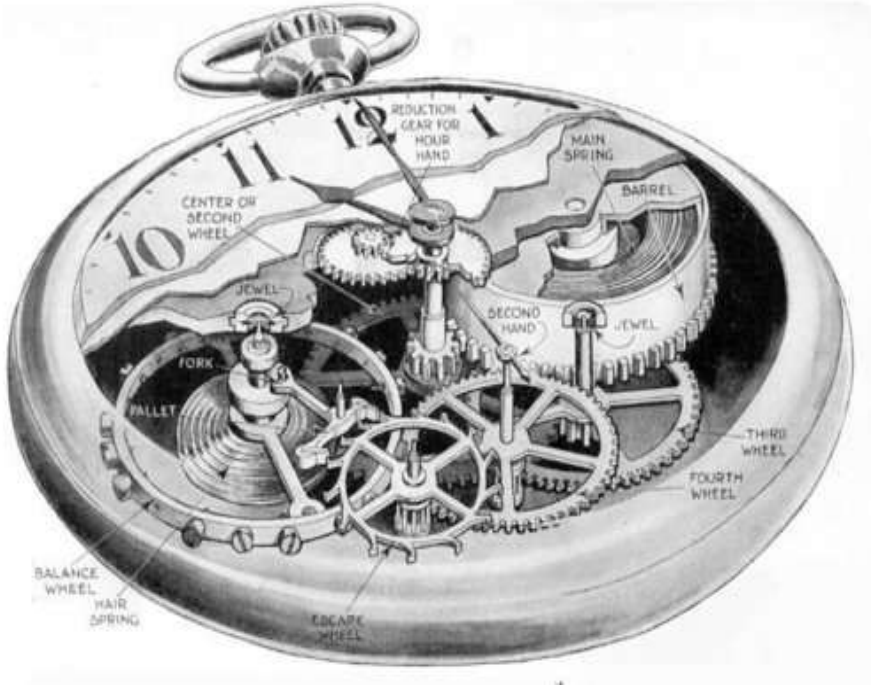
Against Creationism

- Belief that the universe is created by God
- Human beings are special creation in the image of God
- Evidence from revelation and scripture



Against Deism and Argument from Design

- William Paley (1743-1805) *Natural Theology of the Existence and Attributes of the Deity*
- If you found a watch on the beach, you would rationally assume a designer.
- The complexity of nature forces us to rationally assume a designer.



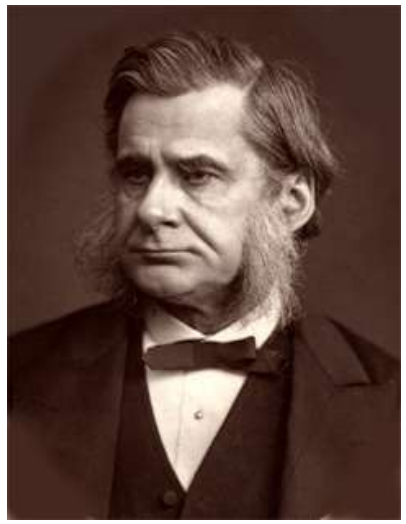
Negative Reactions

I have read your book with more pain than pleasure. Parts of it I admired greatly; parts I laughed at till my sides were almost sore; other parts I read with absolute sorrow, because I think them utterly false and grievously mischievous.



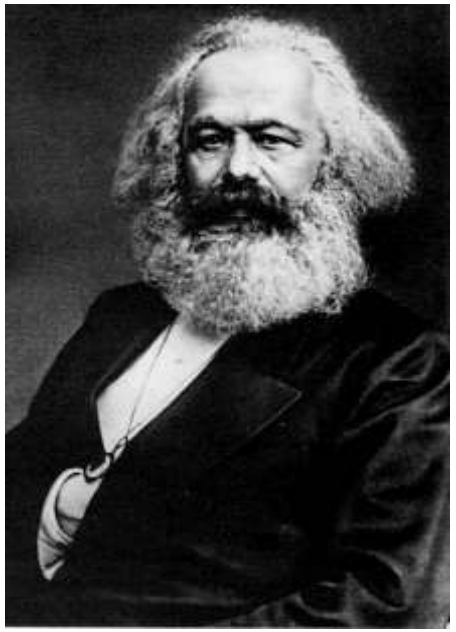
Adam Sedgwick
(1785-1873)
English geologist

Positive Reactions



Thomas Huxley
(1825-1895), English biologist, known as "Darwin's Bulldog".

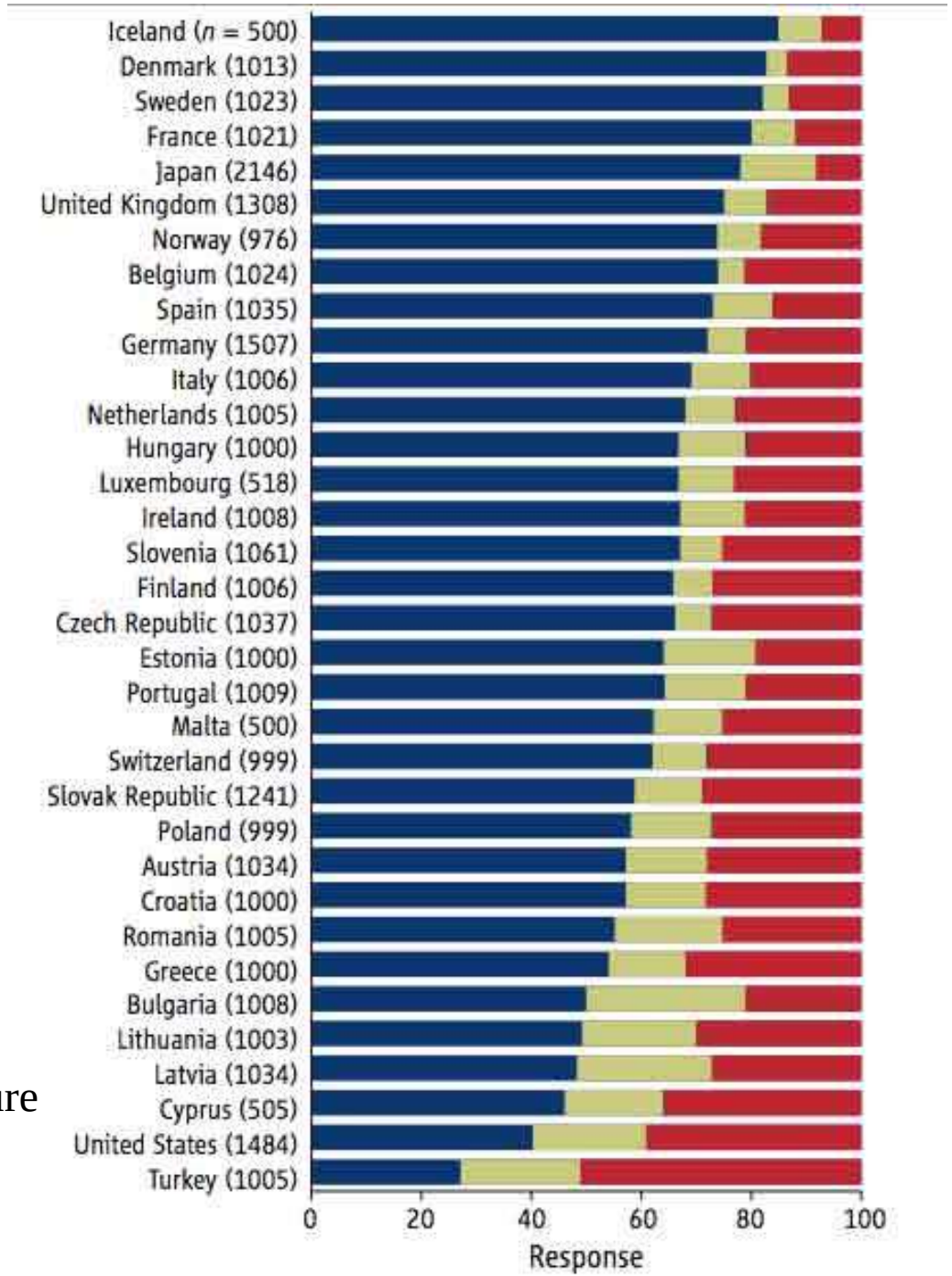
How extremely stupid not to have thought of that!



Karl Marx
(1818-1883), German political theorist

It is remarkable how Darwin rediscovers, among the beasts and the plants, the society of England with its division of labor, competition, opening of new markets, “inventions” and Malthusian “struggle for existence”.

Questionnaire:
Do You Believe in Evolution?



Evolution Less Accepted in U.S. Than Other Western Countries, Study Finds

http://www.mifami.org/eLibrary/Owen_EvolutionSUS.htm

Do You Believe in Evolution?





Notification

- Text 4 *On the Origin of Species*

- Please create a mind map based on Part I of our course.

  Identify a central theme that differs from the established one.

  - The main topic or theme should be clearly labeled and serve as the starting point for all branches.

  Try to find and illustrate the connections between the texts from Part I.

  - Use arrows or lines to show these relationships, and provide brief annotations where necessary to explain the linkage.

  Branches for Major Topics:

  - Create main branches radiating from the central theme, each representing a key subtopic or idea related to the main theme. These branches should be labeled with a single word or short phrase.

  Sub-branches for Details:

  - Further break down each main branch into additional sub-branches to explore specific details, examples, or related concepts.
  - These sub-branches should provide supporting information that enhances understanding of the main branches.

